

Carpathian Earthship: Shelter, Food Store, Co-working

Carpathian OER Commons — Buildings Module

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Maturity: L0, a marked draft — a concept design drawn to dimensions, with quantities taken from the drawings and priced from dated listings, published so that it can be read and changed in the open. Nothing is engineered: every wall and slab thickness is drawn, not calculated, and the slope, the ground, and the winter sun are assumed, not measured. The costs are an orientation, not an estimate anyone may build from. Read every figure below at this level and no higher. The module moves to L1 when the slope is surveyed, the ground investigated, the structure designed by an engineer, and every line in Section 6 carries a dated cost and its basis.

Governed by Section 2 of the compendium. No child appears in this module in any form. The site is not named; site plans are schematic; photographs show structures and systems only. Design loads are site capacity, used for sizing only. The shelter is described in standing format: what it must do and how it is sized, never where it stands; its own supply is given as a principle and a method, without its figures.

A new module, not converted from a working paper. It is a building type that answers three rows of the component map at once — the shelter, the food store's cellar, and a working room — and one row it does not yet have, a wellness room with a sauna. It is its own row in the Buildings group of the map, decided 2026-10-06: the shelter row keeps the standard, siting, and certification, and the food storage row the keeping and preserving.

Permits and local rules. This module is a guide, used in different regions and countries under different laws and guidelines. Depending on where it is built, at what scale, and for whom, permits, registrations, or approvals may be required — for building, water, sanitation, fire, waste, food, or the use of forest and land. Section 10 records what was found for Ukraine, dated, and unchecked until signed; a reader elsewhere checks their own rules with the competent authorities before building, and nothing here replaces that.

1. Purpose

One building dug into a south-facing slope, with the turf carried on over its roof, that holds a protective shelter for 24 to 30 people, a cold food store, a winter garden at the entrance with a sauna and a relax area, and a co-working room for 8 to 10 desks on the floor above. It is for a mountain site that needs all four and would rather keep its slope than build four buildings on it.

Six rules. The slope stays: the building goes into the hill, and the turf runs on over both roofs, so that from above only a chimney, a flue, vents, an air intake, and a hatch show. Water out first: the drain, the membrane, and the falls are built before anything that they protect. Warm, cold, and protected kept apart: insulated walls between the warm rooms and the food store, and the shelter in its own box behind a wall a metre thick, under undisturbed earth, with nothing built above it. No straight line from glass to shelter:

the way in is through a lobby with two offset doors. Every door passes a person 1.80 m tall: at least 0.9 by 2.0 m clear on every main route, round doors included. Concrete where the earth pushes, stone and timber where people look.

What is taken, and what is changed. The name and three ideas come from the Earthship of Michael Reynolds and Earthship Biotecture: south glazing as a buffer, earth as thermal mass, and the building's systems designed together. Changed here: no tyres, because the shelter and the food store sit against the earth walls; reinforced concrete where the earth pushes, because the slope is steep, wet, and loaded with snow; wood, not the sun, as the winter heat, because Carpathian winters are cloudy; and drainage before storage, because the problem here is too much water, not too little. The name is kept, decided 2026-10-06, with its source named here.

Connections. Depends on: a slope survey and a ground investigation; an engineer, for the structure and the shelter; the wood heat module, for the co-working stove, the sauna stove, their chimneys, and the hot water; the energy module, for the shelter's fan and light on the essential circuit; the water module, for supply; the greywater module, for the showers and the plunge; the toilets module, for whether the shelter's WCs are flush or terra preta toilets; the insulation module; the written enquiries in Section 10. Feeds: the shelter row, with its structure; the food storage row, with its cellar; the washroom row, with two shower rooms; the adult strand, with a working room.

Patterns. Grows from: not yet named. How homesteads here kept their cellars and stores, whether a building dug into a slope has a local form, and where a sauna or bath stood are questions for the elders [for the elders]. Feeds: not yet named. Resonates with: 168 Connection to the Earth — the building meets the slope by going into it; 169 Terraced Slope — the two floors step up the hill; 197 Thick Walls; 128 Indoor Sunlight — the most-used rooms behind the south glass; 161 Sunny Place — the relax area in the winter garden; 145 Bulk Storage — the pantry and the root cellar; 230 Radiant Heat — the masonry stove. In tension with: 104 Site Repair — the cut takes deep, undisturbed ground to keep the surface whole; the turf comes back, the ground below does not. The walk status of each is set after the Phase 1 walk, on a later date; until then, unknown (compendium Section 26).

2. Context of validity

Condition	Range this design holds for
Climate	design_temperature −20 °C; heating_season
Snow	[set], from the loads standard (Section 10)
Altitude	[set]
Slope	about 30°, [assumed]; 15° to 35° by adaptation

Condition	Range this design holds for
Aspect	south to south-east, winter sun on the glass
Ground	firm, drained, no slide history, [set]
Seismic	[set], from the seismic standard (Section 10)
Shelter	24 to 30 people, a design load
Co-working	8 to 10 desks
Season of use	all year
Water, power	from the water and energy modules

The design is drawn for a slope of about 30° behind a cut bank about 2 m high; the drawings and every quantity below follow from that. Everything that makes this building expensive — the depth of the cut, the cover over the shelter, the height of the escape shaft — scales with the slope, so a measured slope changes the whole module.

3. Sizing method

What you must measure yourself before using this module:

- **Your slope** — a levelled profile from the yard to at least 25 m up the hill along the building's axis, and two more at its sides; by a surveyor, or with a level and a staff.
- **Your ground** — trial pits or boreholes to below the deepest floor, with the soil, the rock, and the water in them recorded; a geotechnical engineer reads them, including the risk of slides.
- **Your winter sun** — the horizon in December from the position of the glass, at midday and two hours either side.
- **Your width** — the width of level ground you can open, with a metre of working space at each side.
- **Your radon** — a two-month test in any existing cellar nearby, and again in the building before it is lived in.

3.1 The slope sets the building

The ground behind the bank is taken as rising 0.6 m for every metre back, about 31°, [assumed]; the floor of the lower floor and the shelter is at the yard's level.

Quantity	Calculation	Our example
Ground over a point y m back	$2.0 + 0.6 \times y$	—
Earth over the shelter	ground at y 14.5 – roof at 3.6	about 7 m
Back face of the cut	ground at y 19.8 + 0.4 m	about 14.3 m
Escape shaft, floor to hatch	ground at y 19.8 – 3.0 m	about 10.9 m
Excavation, bank volume	Section 3.3	about 5,200 m ³

This is the finding of the draft. On a slope this steep, each metre the shelter goes back into the hill adds 0.6 m to the cut, the cover, and the shaft. A 7 m cover is more than a protective structure may need, and a 14 m back cut cannot stand open in most ground without support. The shelter can sit further forward, under less cover, with a shorter shaft; the escape route can leave sideways onto the slope instead of climbing a shaft; or both. The engineer's cover and the measured slope decide it, [set].

3.2 The shelter

Quantity	Calculation	Our example
Floor area, inside	14.0×7.0 m	98 m ²
Floor area a person	$98 \div 30$ people	about 3.3 m ²
Volume a person	98×3.0 m $\div$ 30	about 9.8 m ³
Clear height	as drawn	3.0 m
Bunks, two tiers	$30 \div 2 = 15$; drawn 16	16, 32 berths
Bunks, three tiers	$30 \div 3$, by the norm	10 or 11
Air, clean ventilation	the norm's rate a person	[held]

The norm. ДБН B.2.2-5:2023 asks at least 2.5 m of clear height in new construction (clause 7.1.5), and two-tier bunks up to 2.9 m and three-tier above it (clause 7.1.6), as read on 2026-10-06. At 3.0 m the drawings' two-tier bunks fall under the three-tier rule: the engineer either lowers the room to 2.9 m or fits three tiers, which frees about a third of the bunk floor. The minimum area and volume a person — 0.5 m² with bunks and 1.5 m³ — are taken from a secondary source, a conference paper, and are read in the norm before they are relied on; the design gives about six times the area, because it is meant for days, not hours.

The supply. Stored water, light, the fan's power, and sanitation for the people the shelter holds are sized from the norm and from the energy module's essential circuit, and are kept in the workshop (compendium Section 2). The method: people $\times$ the norm's daily rate $\times$ the days the norm sets; the fan by the norm's rate a person in each mode, with a hand drive.

Ways out. The way in, through the hall and a lobby with two protective doors that are never in line; a second lobby into the root cellar, so that food is reached without leaving; and an escape route through the north wall, by a short tunnel to a shaft with a ladder and a hatch that opens from inside. The norm's requirements for exits, the hatch, and its distance from buildings are read before the route is fixed, [set].

3.3 Quantities from the drawings

Every thickness below is drawn, not designed: the engineer replaces it, and the quantities with it.

Element	As drawn	Concrete, m ³
Lower floor slab	0.3 m	50.5
Lower floor walls, side and front	0.6 and 0.4 m	53.9
Lower roof, upper floor slab	0.5 m	84.2
Upper floor walls, knee wall, roof	0.6, 0.4, 0.3 m	78.1
Wing walls	0.4 m	7.6
Separation wall, less the lobbies	1.0 m	47.2
Shelter floor, roof	0.4 and 0.6 m	128.0
Shelter walls	1.1, 1.3, 0.8 m	88.5
Escape tunnel and shaft	0.3 m	18.4
Footings and thickenings	10 percent	55.6
Total		612

Quantity	Calculation	Our example
Rebar, general	302 m ³ $\times$ 90–110 kg, [assumed]	27–33 t
Rebar, protective	310 m ³ $\times$ 120–150 kg, [assumed]	37–47 t
Excavation, main	18.4 $\times$ f depth along 19.8 m	about 3,040 m ³

Quantity	Calculation	Our example
Temporary batters, 2:1	back and sides, [assumed]	about 2,150 m ³
Building volume	outside, all parts	about 1,440 m ³
Backfill to the ground	excavation – building	about 3,750 m ³
Membrane, walls and roofs	buried faces	about 590 m ²
Membrane, under slabs	both floors	about 296 m ²
Insulation, 150 mm	warm rooms' buried faces	about 481 m ²
Perimeter drain	back, sides, upper wall, toe	about 82 m
Glazing	winter garden and upper floor	about 45 m ²

The excavated earth is about what the cover, the batters, and the berm take back — the upper floor's roof stands about 3 m above the original ground at its front, and is bermed — so little is left over; any surplus or shortfall is set by the survey, [set].

3.4 Doors

The rule: at least 0.9 m clear width and 2.0 m clear height across that width on every main route, so that a person 1.80 m tall, a wheelchair, or a trolley of crates passes; 0.8 m for the sauna. A round door is a circle of radius r whose centre stands c above the floor, so that the floor cuts it:

Quantity	Calculation	D1, r 1.25, c 1.1	D2, D3, r 1.15, c 1.0
Height	$c + r$	2.35 m	2.15 m
Width at the floor	$2 \times \sqrt{(r^2 - c^2)}$	1.19 m	1.14 m
Width at 2.0 m	$2 \times \sqrt{(r^2 - (2.0 - c)^2)}$	1.73 m	1.14 m
0.9 × 2.0 m box, spare	$r - \text{corner distance}$	62 mm	53 mm

A plain circle standing on the floor is too narrow at the feet; the cut circle is what lets a round door pass people. The spare is at the structural opening, for the frame and the rebate. In the first sketches D1 was 2.4 m across, with 12 mm to spare, and D2 and D3 were 2.2 m, with 3 mm; they are 2.5 m and 2.3 m here for that reason, decided 2026-10-06. The door schedule, sheet 4, draws every door with a person 1.80 m tall in it.

3.5 Heat, light, and the cold

The winter garden is an unheated buffer between the yard and the hall. The co-working room and the wellness room are heated by the wood heat module's appliances: a masonry stove on the co-working room's back wall and a wood sauna stove fired from the utility room, each with its own flue; their heat loads are calculated when the module's buildings are, [set]. The food store keeps cold by the earth behind it and two vents, behind an insulated wall; the pantry is held at 8 to 14 °C and the root cellar at 1 to 5 °C, [set], as targets for the food storage module to confirm.

4. Bill of materials

Quantities from Section 3.3, for the drawn example. Alternatives are named in the design; none is yet known to work here.

Item	Quantity	Alternatives in the design
Concrete C25/30, hydrophobic	about 612 m ³	less, by the engineer
Rebar A500C	about 64–80 t	none named
Bitumen membrane, two layers	about 590 m ²	PVC membrane
Membrane under the slabs	about 296 m ²	none named
Protection and drainage board	about 590 m ²	none named
XPS insulation, 150 mm	about 481 m ²	foam glass gravel
Perimeter drain, pipe, gravel	about 82 m	none named
Timber glazing, triple	about 45 m ²	double, low-e
Round timber doors D1, D2, D3	3	arched doors
Entrance door D10	1	none named
Protective-hermetic doors	5	none named
Protective escape hatch	1	none named
Filter-ventilation set, hand drive	1	none named
Bunks, shelter	10–16	none named
Inner doors, glass and timber	11	none named

Item	Quantity	Alternatives in the design
Insulated cold-store doors	2	none named
WCs, basins, showers	3, 3, 4	terra preta toilets
Sauna lining, benches	about 23 m ²	none named
Stone for facing and wing walls	about 39 m ²	concrete, rendered
Sod and topsoil, re-laid	the cut's own	none named

The stoves, the chimneys, and the water heating are in the wood heat module; the shelter's fan power and light in the energy module.

5. Build

Step	Trade	Time	If done out of order
Survey, trial pits, ground report	engineers	[from L2]	wrong cut
Structure and shelter designed	engineer	[from L2]	[from L2]
Enquiries, permit	household	[from L2]	stop order
Drain and cut support first	groundworks	[from L2]	slip
Excavation in benches	groundworks	[from L2]	slip
Shelter floor, walls, roof	concrete	[from L2]	[from L2]
Separation wall, lower floor	concrete	[from L2]	joints leak
Upper floor walls and roof	concrete	[from L2]	[from L2]
Membrane, board, insulation	waterproofer	[from L2]	leaks
Backfill in layers, cured walls	groundworks	[from L2]	cracks
Sod and topsoil re-laid	landscaper	[from L2]	erosion
Glazing and doors	joiner	[from L2]	broken glass
Protective doors, hatch, fan	specialist	[from L2]	[from L2]

Step	Trade	Time	If done out of order
Interiors, wellness, food store	fit-out	[from L2]	[from L2]
Shelter tested sealed, air test	engineer	[from L2]	[from L2]

Backfill is placed only against walls that have cured and only after the membrane is checked: a leak found after backfill is a leak for good. The back cut is never opened to its full height before its support is in.

6. Cost

An orientation, not an estimate. Prices read on 2026-10-06 from listings in western Ukraine, national online shops, and published tenders; each line below names its basis, and the sources are in Section 12. Pages dated 2025 to 2026, or undated and read on that day. Place: western Ukraine, not a quotation for the site. Exchange rate: exchange_rate, 52 UAH to the euro.

Item	UAH thousand	EUR thousand	Basis
Machine hours, 179	179–268	3.4–5.2	rate quoted, output assumed
Perimeter drain, 82 m	75–101	1.4–1.9	quoted, pipe assumed
Cut support	[set]	[set]	engineer
Earth surplus or shortfall	[set]	[set]	by the survey
Concrete, 612 m ³	2,723–2,980	52–57	quoted
Rebar, 64–80 t	3,123–4,014	60–77	quoted, kg assumed
Formwork, fixing, pouring	2,326–4,590	45–88	quoted
Concrete pump, 100 h	190–600	3.7–11.5	quoted, hours assumed
Delivery up the mountain	[set]	[set]	by quotation
Membrane, walls and roofs	531–700	10.2–13.5	quoted, 2 layers
Membrane under the slabs	133–176	2.6–3.4	quoted, one layer
Protection board	[set]	[set]	not priced

Item	UAH thousand	EUR thousand	Basis
XPS, 150 mm	287–542	5.5–10.4	quoted, scaled
Glazing, supply	664–1,300	12.8–25.0	quoted
Glazing, fitting	[set]	[set]	not found
Stone laying, 26 mason-days	52–65	1.0–1.2	wage, rate assumed
Round doors D1, D2, D3	49–570	0.9–11.0	bracket only
Entrance door D10	16–190	0.3–3.7	bracket only
Inner and glass doors	38–68	0.7–1.3	quoted, proxy
Cold-store doors D7	[set]	[set]	not found
Protective doors, 5	308–732	5.9–14.1	quoted
Protective hatch	53	1.0	quoted, one listing
Filter-ventilation set	[set]	[set]	see below
Shelter bunks	[set]	[set]	not priced
Shelter interior, 98 m ²	882–1,372	17.0–26.4	quoted
Warm rooms finished, 136 m ²	1,224–2,584	23.5–49.7	quoted
Winter garden, 35 m ²	318–495	6.1–9.5	quoted
Food store shelving, finish	[set]	[set]	not priced
WCs, basins, showers	33–113	0.6–2.2	quoted
Fitting them, 4 rooms	14–37	0.3–0.7	quoted
Sauna lining, 23 m ²	10–27	0.2–0.5	quoted
Sauna benches, barrier	[set]	[set]	not priced
Subtotal, priced lines	13,227–21,576	254–415	
Design check, 10 percent	1,323–2,158	25–41	assumed

Item	UAH thousand	EUR thousand	Basis
Contingency, 20 percent	2,910–4,747	56–91	assumed
Orientation total	17,460–28,481	336–548	

What the total leaves out. Every [set] line: the support of the back cut, which may be the largest of them; the filter-ventilation set, for which no price was found, and one 2026 tender budgeted about 569 thousand UAH a set on average for ten sets with their valves and blast devices; delivery of concrete up the mountain; the protection board; fitting the glazing; the bunks; the food store's fittings; the sauna's benches; the permit, the survey, and the ground investigation. The two bracketed door lines are not like-for-like prices: the round doors are made to order, and a joiner's quotation replaces the bracket. The wide ranges are the listings' own: the labour for monolithic work, for one, runs from 3,800 to 7,500 UAH a m³. Where the template asks for percentages, they are taken on the priced lines only, and say nothing about the unpriced ones.

Carried by the wood heat module, not in the total. A masonry stove for the co-working room, labour and materials, 40–80 thousand UAH; a wood sauna stove of Ukrainian make, 8.6–27.0 thousand; two chimneys and flues, [set].

How the total compares. About 46 to 74 thousand UAH a m² of gross floor, about 384 m² — two to three times the Ministry's indicative cost of housing in the region for January 2026, about 23 thousand UAH a m²; the priced lines only, so a floor, not a figure. The shelter's own share — its concrete, steel, and placing, the protective works, the membrane on its faces, its part of the earthworks, and the same two percentages, as calculated in the cost model — comes to about 8.1 to 12.1 million UAH, or 270 to 400 thousand a place at 30 people. Public shelters reported in 2025 cost about 64 to 435 thousand UAH a place, about 202 thousand on average, and underground buildings of the same kind 230 to 380 thousand. This one gives each person about six times the minimum floor, which is part of the difference, [assumed].

Where the money is. Concrete, steel, and the work of placing them are about three-fifths of the priced lines. The levers, for the engineer to test: the shelter further forward under less cover; the escape route sideways onto the slope instead of up a shaft; three-tier bunks in a smaller shelter, or a lower room; the upper floor framed in timber with only its back wall in concrete; and, where no shelter is needed, the protective works and about half the concrete gone.

Contribution, not in the total. Stone for the facing and the wing walls, from the site; the sod and topsoil lifted from the cut and laid back on the roofs; the excavated earth, placed back as cover and berm; each with the landowner's leave. Shown at the price of the same bought, [set]. The work is paid.

Running cost.

- The shelter's seals, doors, hatch, and fan checked and serviced, [set].
- The shelter kept dry: ventilation and, if needed, a dehumidifier on the energy module's general circuit, [set].
- The drains flushed and the roof turf kept, [set] hours.
- Firewood for the stove and the sauna, in the wood heat module.

7. Operation and maintenance

Responsible: one named adult for the building and one for the shelter, each with a deputy — [from L2].

What the design already asks:

- **Monthly:** every protective door closed and opened, its seal looked at; the hatch opened from inside; the fan turned by hand.
- **Before each winter:** the drains and their outlets clear; the vents of the food store open and screened; the glazing's seals sound.
- **After heavy snow:** the glazing clear and the top vents free; the hatch and the air intake not buried.
- **Each spring:** the turf on the roofs checked for wet patches, which are the first sign of a failed membrane below.
- **Each year:** the shelter's readiness checked as the rules for protective structures set, [set]; the radon test repeated.

8. Failure modes

From the design, not from experience.

Failure	Early sign	Repair and cost
Membrane leak	damp patch, white salts	dig out; [from L2]
Cut slips during the dig	cracks at the top	stop; support
Drain blocked	wet back wall, spring seep	flush; [from L2]
Condensation on the glass	streaming panes	vents, fan
Food store too warm	above 5 °C in the cellar	vents; insulation
Stale air, shelter sealed	headache, CO2 reading	fan by hand

Failure	Early sign	Repair and cost
Hatch buried by snow	will not lift	clear before winter
Frost heave at the front	cracked toe, door binds	deeper footing
Radon in the shelter	test above the limit	sealing, venting

9. What we got wrong

Nothing yet: this module has not been built.

10. Regulatory notes

Country	Instrument, number, date	Checked by, date
Ukraine	found 2026-10-06, not checked — see below	—
Romania	not checked	—
Poland	not checked	—
Slovakia	not checked	—
Hungary	not checked	—
Serbia	not checked	—
Czechia	not checked	—

Ukraine, found and not checked. Read on 2026-10-06 on the state register of building norms and on secondary copies; each is read in full and signed before this row counts as checked.

- **ДБН В.2.2-5:2023, Захисні споруди цивільного захисту**, approved by Order No. 702 of the Ministry for Communities, Territories and Infrastructure, 10.08.2023, in force 01.11.2023, with Amendment 1, in force 01.11.2023, and Amendment 2, Order No. 1321 of 13.11.2024, in force 01.04.2025; a consolidated edition of 15.04.2026 is on the register. It names three kinds of protective structure: сховище, протирадіаційне укриття, and споруда подвійного призначення. Clauses 7.1.5 and 7.1.6 read as in Section 3.2. The area and volume a person, the exits, the hatch, the water, the sanitation, the ventilation, and the protection class were not read in the norm.

- **ДБН В.2.6-31:2021**, thermal insulation, as cited in the wood heat module, for the warm rooms' envelope.
- **ДБН В.1.2-2:2006**, loads and actions, for the snow on the roofs and the earth on them; not read.
- **ДБН В.1.1-12:2014**, construction in seismic regions; not read.
- **Law of Ukraine No. 3038-VI of 17.02.2011**, on the regulation of urban development: the permit route and the class of consequences, which a protective structure may raise; not read.
- **The fire rules**, for the sauna stove and the masonry stove, as the wood heat module records them.

The written enquiries: to the regional office of the State Emergency Service, which kind of protective structure a private building of this size is, whether it is entered in the fund of protective structures, and what it must then meet and allow; to the architecture and urban planning authority, the permit route and the class of consequences; to a geotechnical engineer, the ground and the slope; to the makers of protective doors and filter-ventilation sets, certified units and their sizes.

11. Adaptation notes

The design is a seed, not a copy: a different place grows a different form from it.

- **A gentler slope:** less cover over the shelter; the berm built up over it, and the upper floor set further back.
- **A steeper slope:** a deeper cut for the same floor; the shelter forward and the escape route out to the side; a shorter building.
- **Rock near the surface:** less support, more breaking; the shelter where the rock lets it, not where the drawing puts it.
- **Wetter ground:** a drainage blanket behind every buried wall, a heavier membrane, and the lowest floor raised.
- **A colder valley:** thicker insulation, less glass on the upper floor, and a larger stove.
- **A smaller group:** a shelter half the width, two-tier bunks in a lower room, the food store reached the same way.
- **No shelter needed:** no separation wall, lobbies, or protective works; a cellar behind the food store instead; about half the concrete.
- **Water for the shelter:** stored as the water module and the norm set, whatever the site's own sources.

12. Provenance and licence

- **Written by:** Michel Garand. [set — others as they choose to be named]

- **From whose knowledge:** the Earthship of Michael Reynolds and Earthship Biotecture, for the idea, as changed in Section 1; ДБН В.2.2-5:2023 on the state register, e-construction.gov.ua, and a conference paper of Zhytomyr Polytechnic, 2025, for the figures a person; for prices, read 2026-10-06: two ready-mix suppliers' price pages in western Ukraine, concrete; Metinvest-SMC, rebar; Rabotniki.ua and Budver, labour; a machine-hire firm's price page, machine hire; Prom.ua and Epicentr K, membrane, insulation, doors, sanitary ware; Gurtom, timber windows; Енергомонтаж ЛВ and Строй-Снабжение, protective doors and hatch; Allo.ua and Rozetka, sauna stove and lining; Obozrevatel and Minfin, 2026, finishing; Prozorro tenders UA-2026-04-07-007770-a, gravel, and UA-2026-06-01-003207-a, filter-ventilation; Zaxid.net, 2026, the Ministry's indicative housing cost; Ekonomichna Pravda with Nashi Groshi, and Glavcom, 2025, the cost of shelters a place, as built for public buildings; Alexander's A Pattern Language, as prompts.
- **Traditional knowledge:** none taken. The questions on cellars, building into a slope, and the bath are in the pattern line, for the elders.
- **Text:** CC BY-SA 4.0
- **Drawings and designs:** four sheets, v0.7, in drawings/buildings/, SVG beside PNG with the script that draws them — earthship_elevations_roof_plan, earthship_floor_plans_section, earthship_wellness_area, earthship_door_schedule; the cost model and its lines in sources/buildings/; [hardware licence — open decision]
- **Measurement data:** none yet
- **Gate 1:** household decision, 2026-10-07, recorded by Michel Garand — v0.1 may be published as it stands, as a marked draft
- **Wartime review:** 2026-10-07, Michel Garand — v0.1 read against compendium Section 2; nothing places the site; the shelter in standing format, its supply figures out of the drawings; the concept image on photographs of the slope kept out of both repositories; accepted

13. What to check

- **The slope** — surveyed; every quantity in Section 3 recalculated.
- **The ground** — trial pits and a geotechnical report; the slide risk; how the back cut is supported, and its cost.
- **The cover and the escape route** — the engineer's cover for the shelter; whether it moves forward, and whether the route leaves sideways instead of by a shaft.
- **The structure** — every thickness in Section 3.3 designed; the rebar rate in kg a m³ with it.
- **The shelter** — its kind and class from the enquiry; the area and volume a person, the exits, the hatch, the supply, and the air read in ДБН В.2.2-5:2023; two or three tiers of bunks.
- **The winter sun** — the December horizon at the glass.

- **Radon** — tested before and after.
- **Quotations** — every line in Section 6, the [set] lines first: cut support, filter-ventilation set, delivery up the mountain, the round doors by a joiner, glazing fitting, bunks.
- **The exchange rate** — the National Bank's rate on 2026-10-06 was 50.53 UAH to the euro, as read on a secondary copy; exchange_rate is kept at 52 until the site parameters are next reviewed, decided 2026-10-06, so that every module moves together.
- **The round doors** — a joiner's drawing of leaf, frame, and rebate that keeps 0.9×2.0 m clear.
- **The supply figures** — confirmed out of the drawings.
- **The elders** — the three questions in the pattern line.
- **From L2, once built:** build times and out-of-order consequences in Section 5; the maintenance record in Section 7; early signs and repair costs in Section 8; Section 9.

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